

Supplementary Material: NiyamTrace-X

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1 Supplementary Local-T4 Benchmark-Harness Diagnostics

1.1 Expanded Local-T4 Benchmark-Harness Study

We next ran a closure-oriented local experiment on a single NVIDIA T4 to test whether the surrounding tool-agent benchmark harness could be exercised with open-weight models under a fixed, reproducible setup. This study is deliberately separated from the primary NIYAMTRACE-Xevaluation: BFCL, AgentDojo, and the local τ^3 runner call the served model directly through an OpenAI-compatible tool interface. They do *not* route benchmark actions through the NiyamTrace-X semantic authorization layer. The results below therefore characterize model/tool compatibility and the external evaluation harness; they are not direct measurements of NiyamTrace-X safety.

Granite-3.3-2B and Qwen2.5-3B both passed ordinary chat, automatic tool-selection, and forced-tool preflight. BFCL then timed out for both models (process return code 124) without a retained native score. AgentDojo produced nine valid native cases per model across banking, workspace, and travel—three cases per suite—while the Slack slice failed before yielding a valid case. On the six attacked AgentDojo cases, Granite’s native utility/security flags were 1/6 and 0/6; Qwen’s were 1/6 and 1/6. On the three no-injection baselines, Granite recorded utility/security 3/3 and 3/3, whereas Qwen recorded 2/3 and 3/3.

For the local τ^3 runner, Granite completed all six requested trajectories: two airline rewards of 1, followed by two retail and two telecom rewards of 0, for an evaluated mean reward of 0.333. Qwen completed four of six trajectories—two airline and two retail, all with reward 0—while both telecom trajectories exceeded the configured 8,192-token context after reserving output tokens and were not scored. Because the same currently loaded local model also acts as the simulated user in these runs, cross-family τ^3 reward differences are not treated as a controlled head-to-head comparison.

A standalone continuation was intended to fill the missing BFCL/Slack/Qwen-telecom slices and run the independent Hermes-3/Llama

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Table 1. Expanded local-T4 benchmark-harness study. “AD” is AgentDojo. Counts are native evaluable cases/trajectories; NE means not evaluated. These results characterize the external model/tool harness, not the NiyamTrace-X authorization layer.

Checkpoint	Tool preflight	BFCL	AgentDojo	Local τ^3
Granite-3.3-2B	pass	timeout; no score	9 valid across 3 suites; Slack failed 6/6 evaluated; mean 0.333	
Qwen2.5-3B	pass	timeout; no score	9 valid across 3 suites; Slack failed 4/6 evaluated; mean 0.000	
Hermes-3/Llama-3.1-8B-AWQ	not reached	NE	NE	NE

family. It stopped before model execution for all three configured models because the notebook referenced an undefined `HF_TOKEN` variable. We treat that as an infrastructure failure, not as a model failure, and report no Llama benchmark result. Consequently, the planned three-family closure was not evaluated.

2 Supplementary Repository and Runtime Audit Detail

We separately audit the public implementation at <https://github.com/bnssaanirudh/NiyamTrace-X>, pinned to commit `c14661dbd11c42ebd1019b6a1a5c49b8643da137`. The deterministic gate checks target-identity containment, temporal containment, attribute containment, cardinality and approval constraints, and evidence verdicts. One important limitation is explicit in the snapshot: the default evidence path is a stub. We therefore do not use it to support security claims.

The audited 10,000-row prototype corpus also contains substantial duplication. We find 1,668 exact duplicate raw texts and 5,330 rows that duplicate after template normalization, leaving 46.7% of rows unique under that normalization. In a TF-IDF sample audit, 47.7% of sampled rows have a near-duplicate at cosine similarity at least 0.95. These numbers describe the audited prototype corpus, *not* the frozen 2,000-case primary group-disjoint holdout.

In the intended application path, the audited pipeline checks the gate verdict and calls the executor only after an `ALLOW` decision. Its legacy executor API, however, places that responsibility on the caller and does not require a gate-issued capability. In a synthetic in-memory ERP probe, calling the executor directly without a gate decision changes three matching invoices from `OPEN` to `ARCHIVED`. We describe this narrowly as an *internal execution-boundary bypass in the legacy API*; it is not evidence of a remotely exploitable gateway vulnerability.

The audited snapshot also includes a hashable `EffectCertificate`; its `seal()` and `verify()` primitives detect mutation after sealing. The current gate does not yet complete that design: it constructs the certificate without sealing it and uses placeholder values for policy, evidence, and state fields. For that reason, we do not claim cryptographically complete authorization-to-commit binding.